

Rho Sweep - Reference Document

Timeline in: [Regressive Property Taxation with Fairness](#)

Code in: [Github repo](#)

Overview

1. Rho selection

```
python quick_test_models.py \  
  --models linear,lgbm,cov \  
  --rho 3 \  
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \  
  --lgbm-config-key test_best_r2 \  
  --out-dir output/quick_test_best_lgbm_cov \  
  --lgbm-n-jobs 380
```

```
python quick_test_models.py \  
  --rho 3 \  
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \  
  --lgbm-config-key test_best_r2 \  
  --lgbm-n-jobs 1 \  
  --models linear,lgbm,cov \  
  --out-dir output/quick_test_example \  
  --data-path data/CCA0/2025/training_data.parquet \  
  --parallel-models \  
  --sample-frac 1.0
```

```
# not used for the experiments: --early-stopping-rounds 10 \  

```

```
python quick_test_models.py \  
  --rho-range "1e-1,20" \  
  --rho-count 30 \  
  --rho-scale log \  
  --lgbm-hyperparameter-file best_lgbm_baseline_configs.yaml \  
  --lgbm-config-key test_best_r2 \  
  --lgbm-n-jobs 1 \  
  --lgbm-n-estimators 100 \  

```

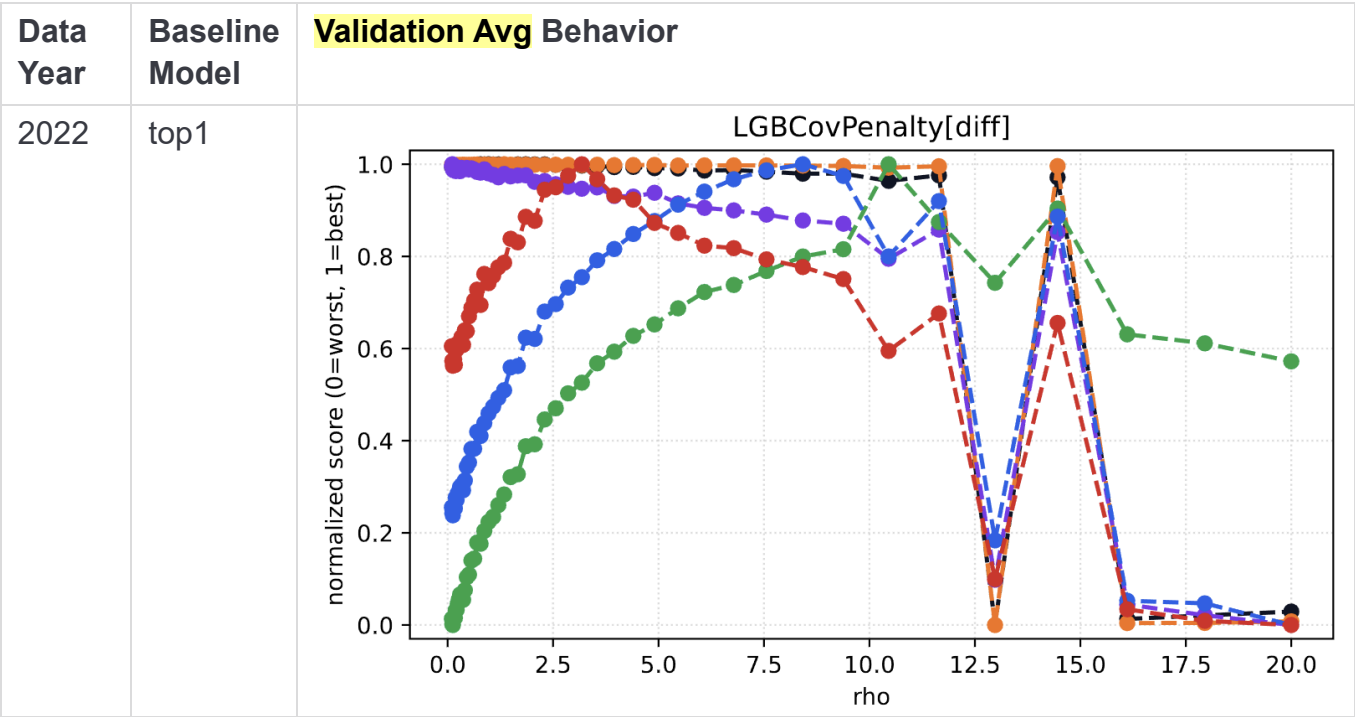
```
--models linear,lgbm,cov \  
--out-dir output/quick_test_example \  
--data-path data/CCA0/2025/training_data.parquet \  
--parallel-models \  
--sample-frac 0.1 \  
--seed 4050 \  
--n-bootstrap-validation 5 \  
--bootstrap-block-freq M \  
--scatter-plot-max-samples 1000
```

Results

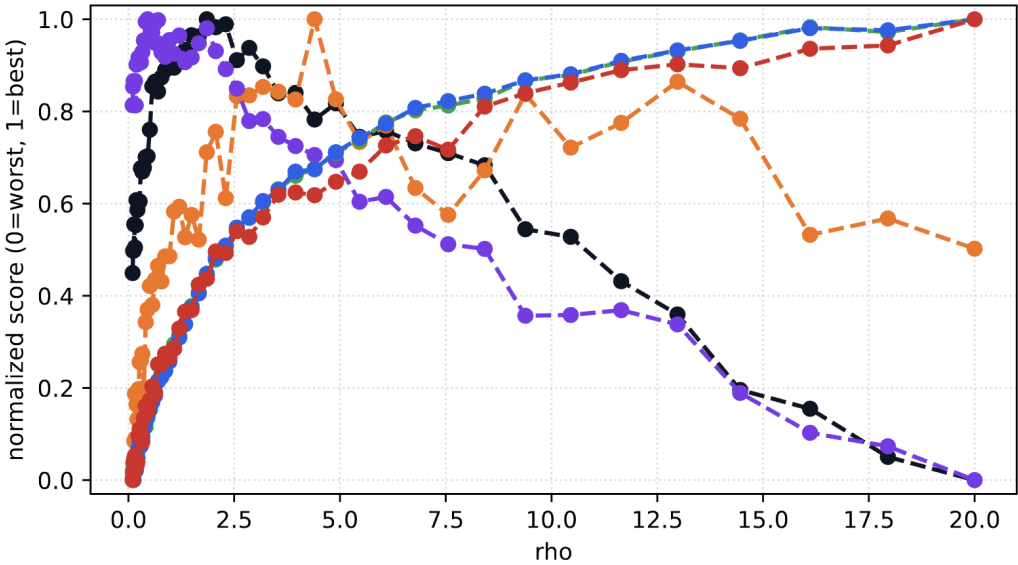
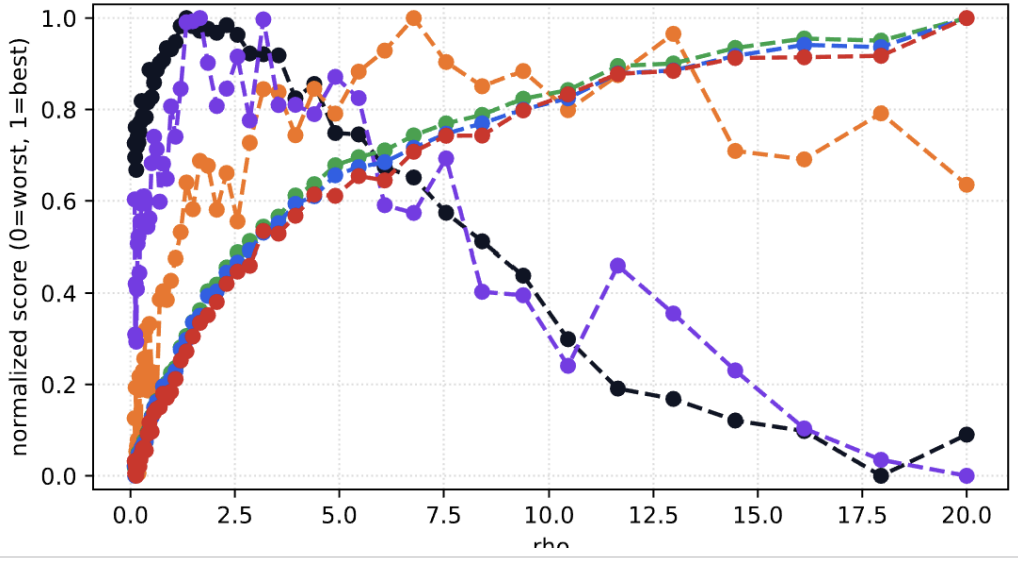
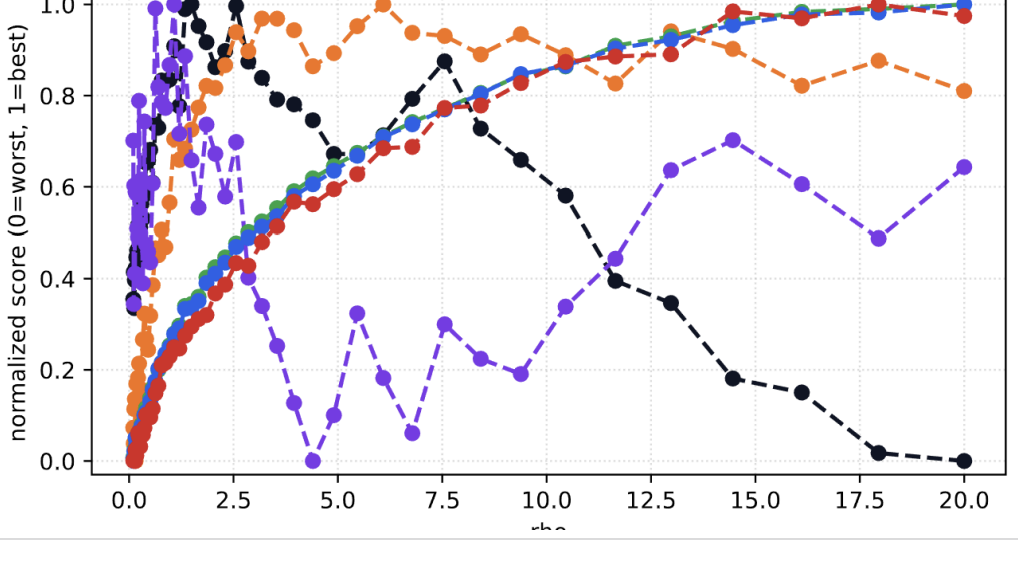
1. All Metrics Evolution w.r.t. ρ

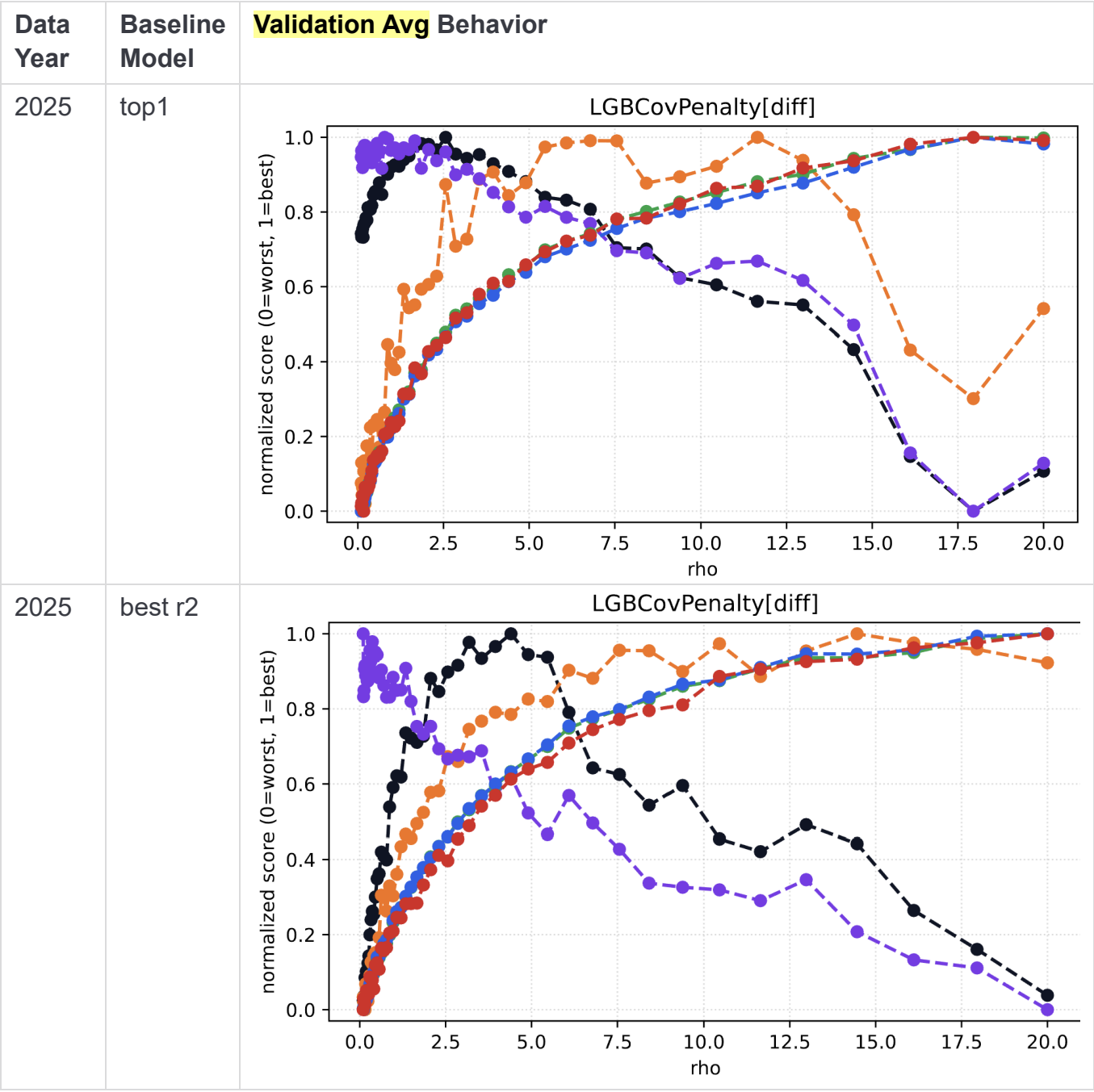
Validation Average

Metrics:



Data Year	Baseline Model	Validation Avg Behavior
2022	best r2	LGBCCovPenalty[diff]
2023	top1	LGBCCovPenalty[diff]

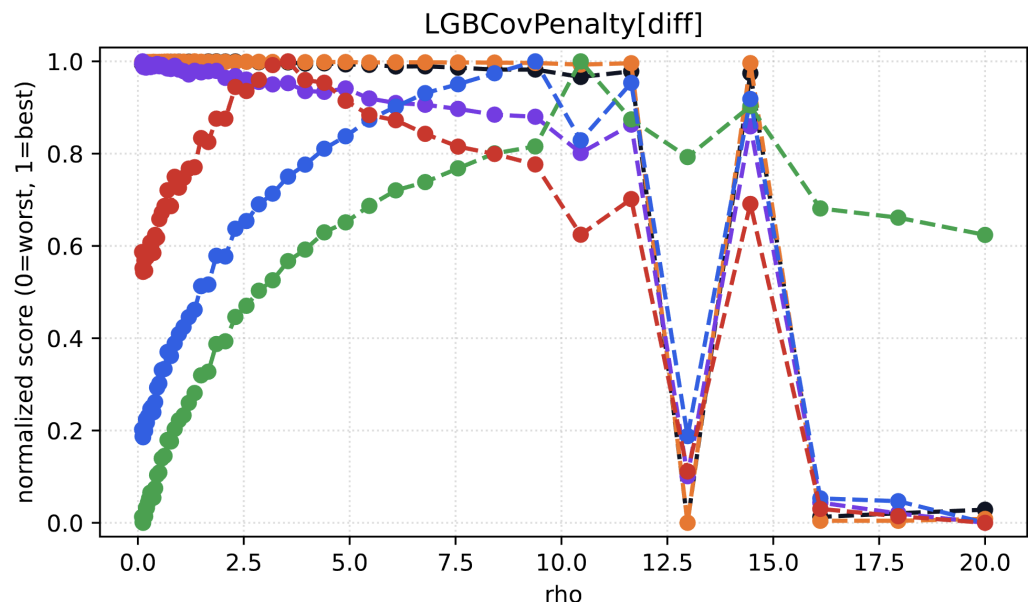
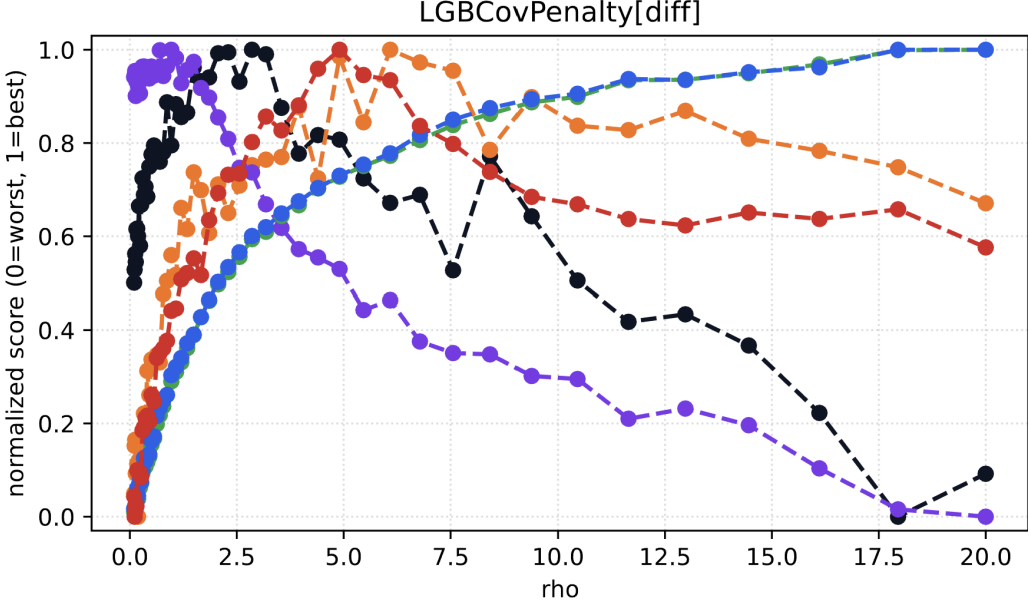
Data Year	Baseline Model	Validation Avg Behavior
2023	best r2	LGBCovPenalty[diff]  This plot shows the performance of several models as a function of the regularization parameter rho. The y-axis represents the normalized score, ranging from 0.0 (worst) to 1.0 (best). The x-axis represents rho, ranging from 0.0 to 20.0. The models are represented by different colored lines and markers: black (solid), orange (dashed), red (solid), green (dashed), blue (solid), and purple (dashed). The black model starts at a high score of approximately 0.45 at rho=0, peaks at 1.0 around rho=2.5, and then declines to 0.0 by rho=20.0. The orange model starts at 0.0, peaks at 1.0 around rho=4.5, and then declines to approximately 0.5 at rho=20.0. The red, green, and blue models all start at 0.0 and increase to reach a score of 1.0 by rho=20.0. The purple model starts at 1.0, peaks at 1.0 around rho=1.5, and then declines to 0.0 by rho=20.0.
2024 (old)	top1	LGBCovPenalty[diff]  This plot shows the performance of several models as a function of the regularization parameter rho. The y-axis represents the normalized score, ranging from 0.0 (worst) to 1.0 (best). The x-axis represents rho, ranging from 0.0 to 20.0. The models are represented by different colored lines and markers: black (solid), orange (dashed), red (solid), green (dashed), blue (solid), and purple (dashed). The black model starts at a high score of approximately 0.65 at rho=0, peaks at 1.0 around rho=2.5, and then declines to approximately 0.1 at rho=20.0. The orange model starts at 0.0, peaks at 1.0 around rho=7.5, and then declines to approximately 0.65 at rho=20.0. The red, green, and blue models all start at 0.0 and increase to reach a score of 1.0 by rho=20.0. The purple model starts at 1.0, peaks at 1.0 around rho=2.5, and then declines to 0.0 by rho=20.0.
2024 (old)	best r2	LGBCovPenalty[diff]  This plot shows the performance of several models as a function of the regularization parameter rho. The y-axis represents the normalized score, ranging from 0.0 (worst) to 1.0 (best). The x-axis represents rho, ranging from 0.0 to 20.0. The models are represented by different colored lines and markers: black (solid), orange (dashed), red (solid), green (dashed), blue (solid), and purple (dashed). The black model starts at a high score of approximately 0.35 at rho=0, peaks at 1.0 around rho=2.5, and then declines to 0.0 by rho=20.0. The orange model starts at 0.0, peaks at 1.0 around rho=7.5, and then declines to approximately 0.8 at rho=20.0. The red, green, and blue models all start at 0.0 and increase to reach a score of 1.0 by rho=20.0. The purple model starts at 1.0, peaks at 1.0 around rho=1.5, and then declines to approximately 0.65 at rho=20.0.

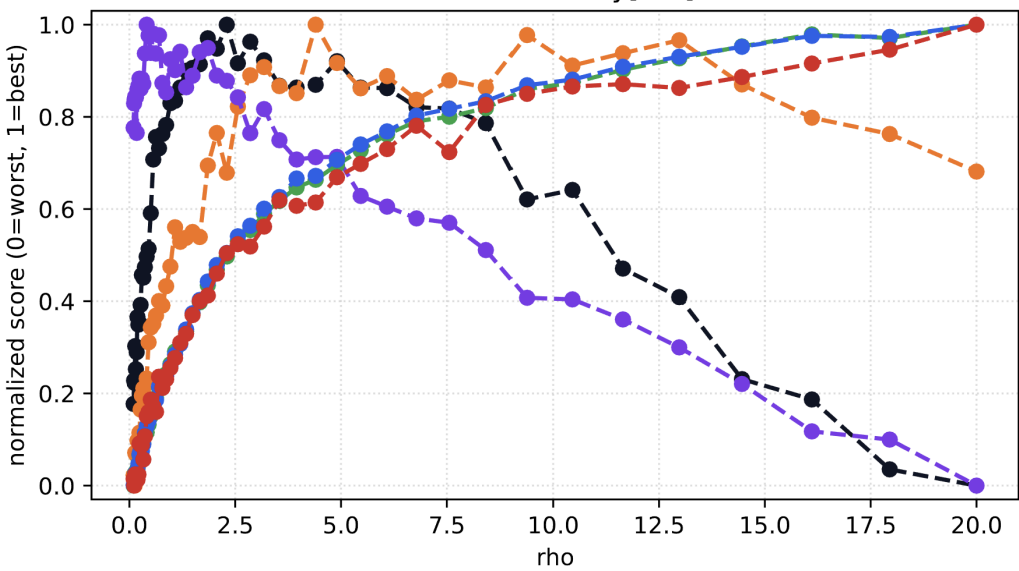
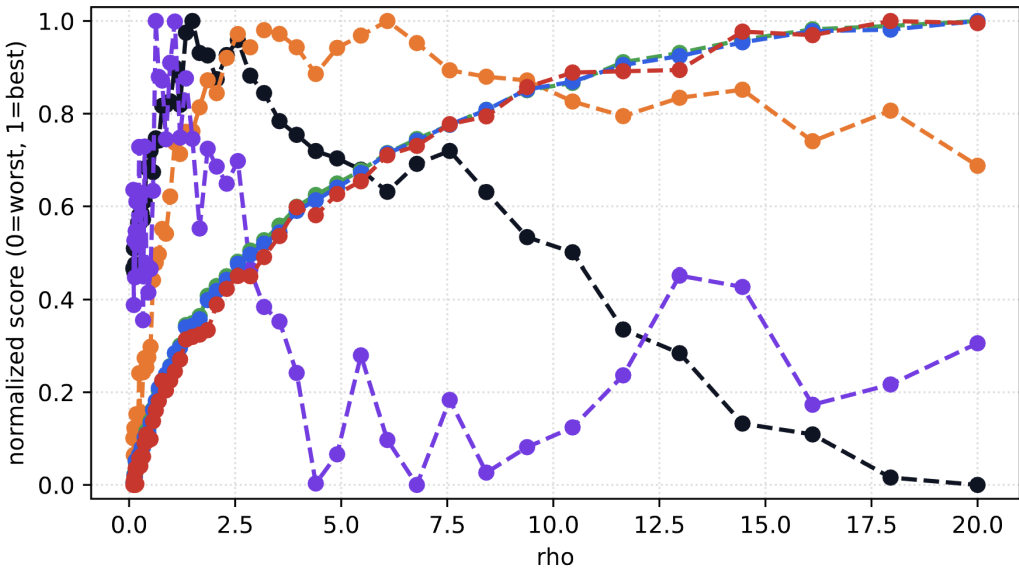


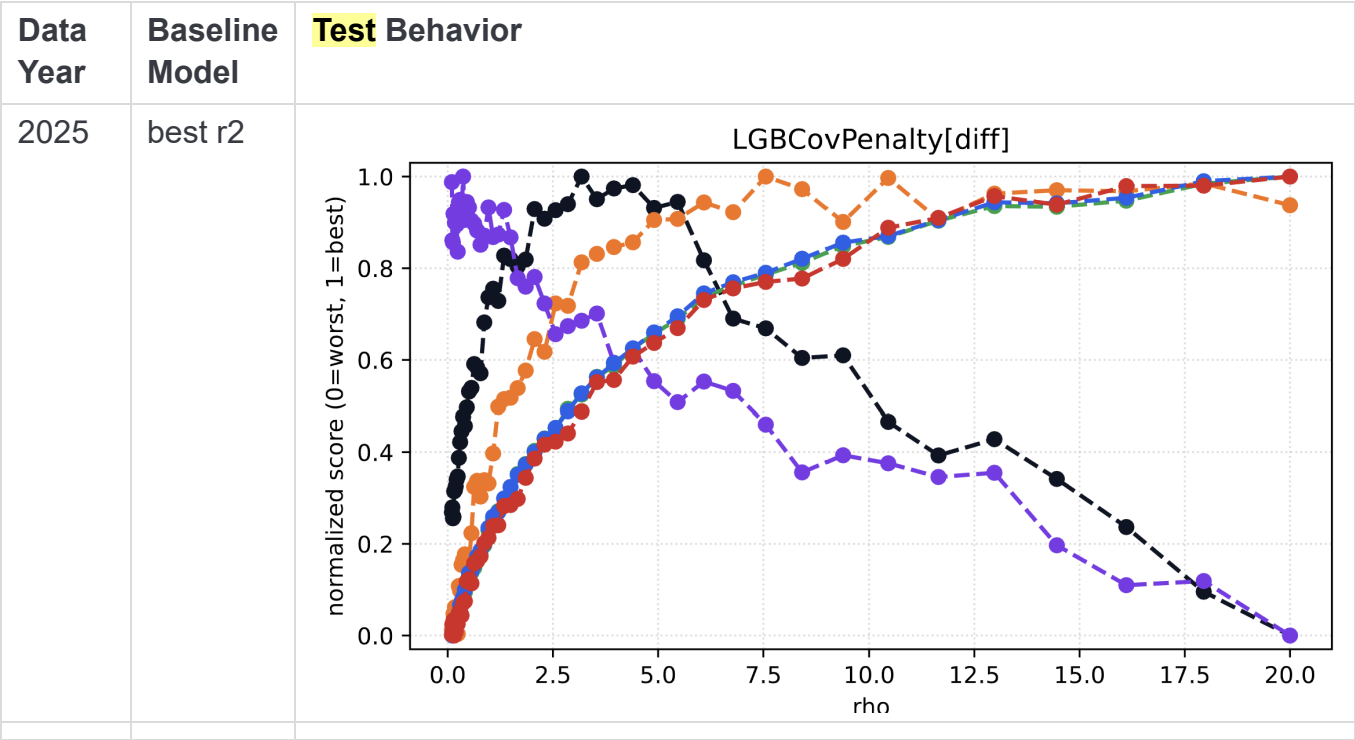
Test Average

Metrics:



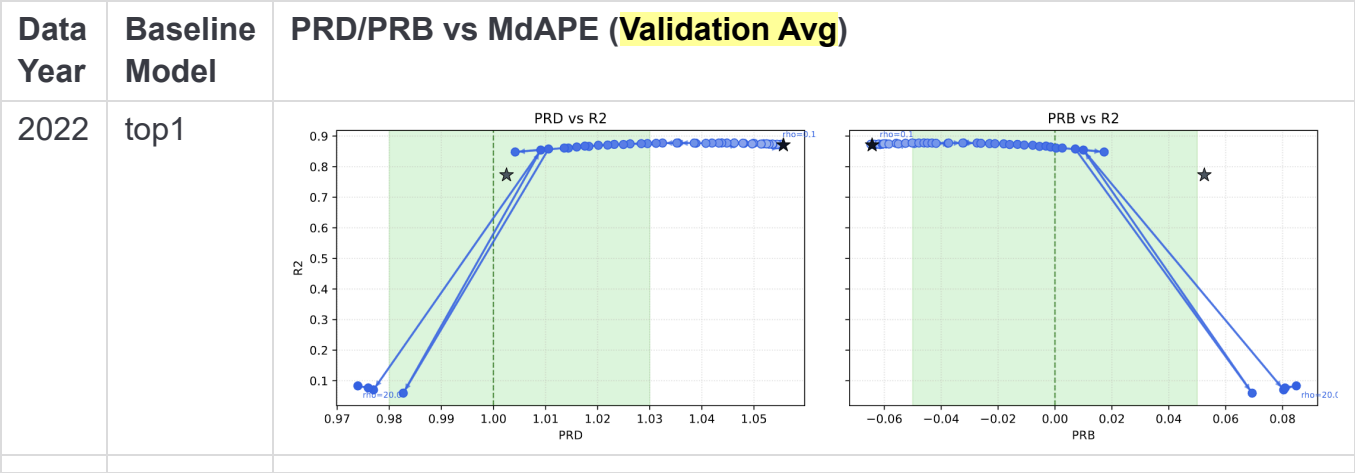
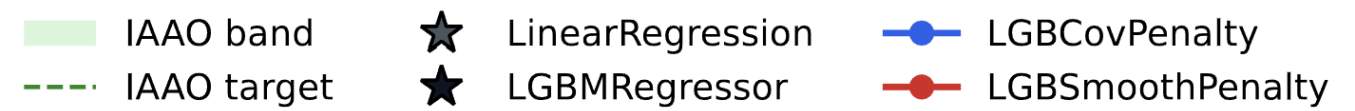
Data Year	Baseline Model	Test Behavior
2022	top1	 Line plot titled "LGBCovPenalty[diff]" showing normalized score (0=worst, 1=best) versus rho (0.0 to 20.0). The plot displays multiple curves (solid and dashed lines) in various colors (blue, green, red, orange, black). The scores generally increase with rho, peaking around rho=10-15, and then drop sharply to near zero at rho=20.0.
2022	best r2	 Line plot titled "LGBCovPenalty[diff]" showing normalized score (0=worst, 1=best) versus rho (0.0 to 20.0). The plot displays multiple curves (solid and dashed lines) in various colors (blue, green, red, orange, black). The scores generally increase with rho, peaking around rho=10-15, and then drop sharply to near zero at rho=20.0.

Data Year	Baseline Model	Test Behavior
2023	best r2	LGBCovPenalty[diff] 
2024 (old)	best r2	LGBCovPenalty[diff] 



2. Tradeoffs

Validation Avg



Data Year	Baseline Model	PRD/PRB vs MdAPE (Validation Avg)
2023	best r2	PRD vs R2 PRB vs R2
2024	best r2	PRD vs R2 PRB vs R2
2025	best r2	PRD vs R2 PRB vs R2

Test

IAAO band

IAAO target

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LinearRegression

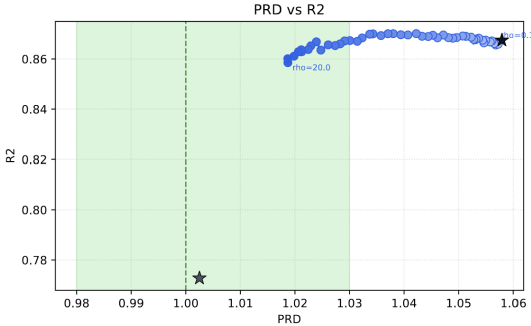
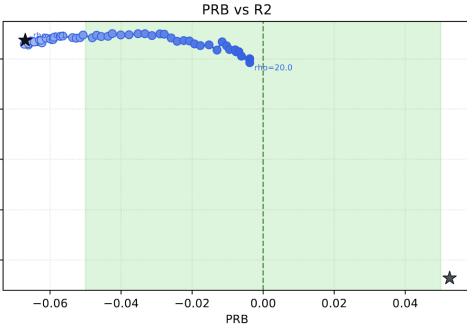
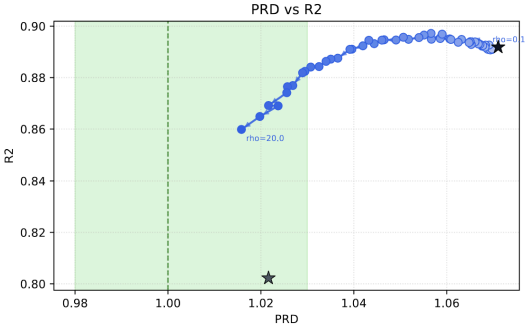
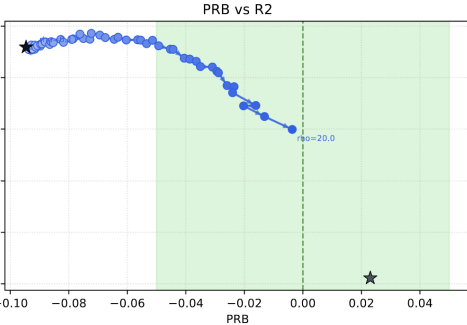
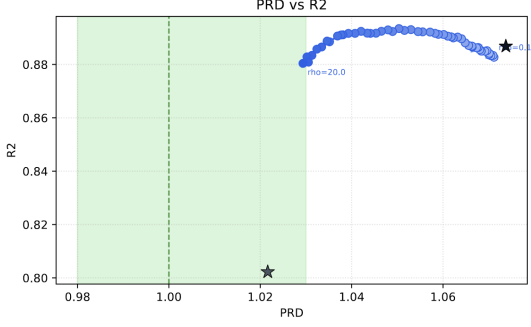
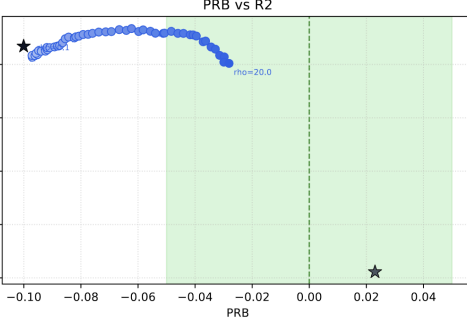
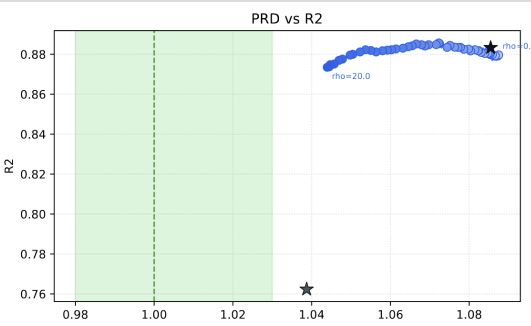
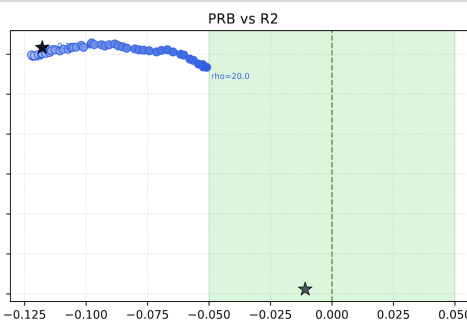
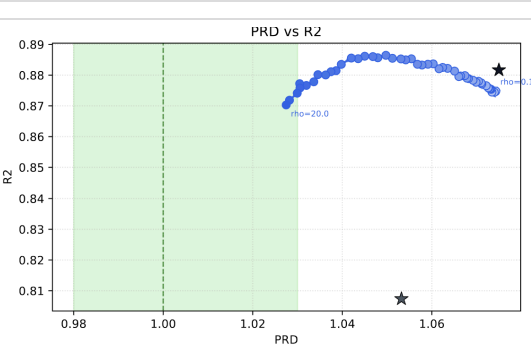
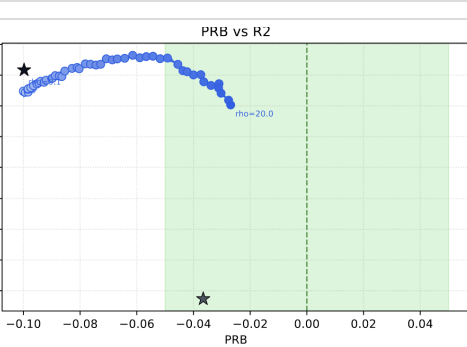
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LGBMRegressor

LGBCovPenalty

LGBSmoothPenalty

Data Year	Baseline Model	PRD/PRB vs MdAPE (Test)
2022	top1	PRD vs R2 PRB vs R2

Data Year	Baseline Model	PRD/PRB vs MdAPE (Test)	
2022	best r2	 	
2023	top1	 	
2023	best r2	 	
2024 (old)	best r2	 	
2025	best r2	 	

Notes

 **Tags?** >

#RA #MIT #RegressivePropertyTaxation